Practical No – 8

Managing Data Tables and Spatial Data Sets: Table joins, spatial joins, points in polygon analysis, performing spatial queries

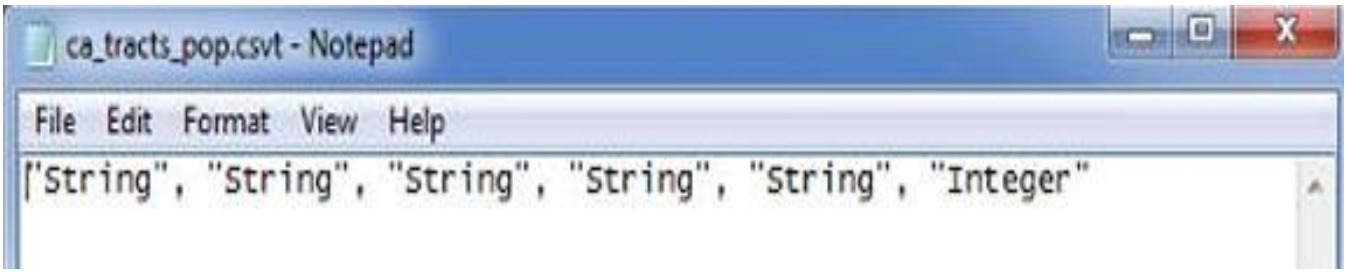
❖ Managing Data Tables and Spatial Data Sets:

A. Table Joins

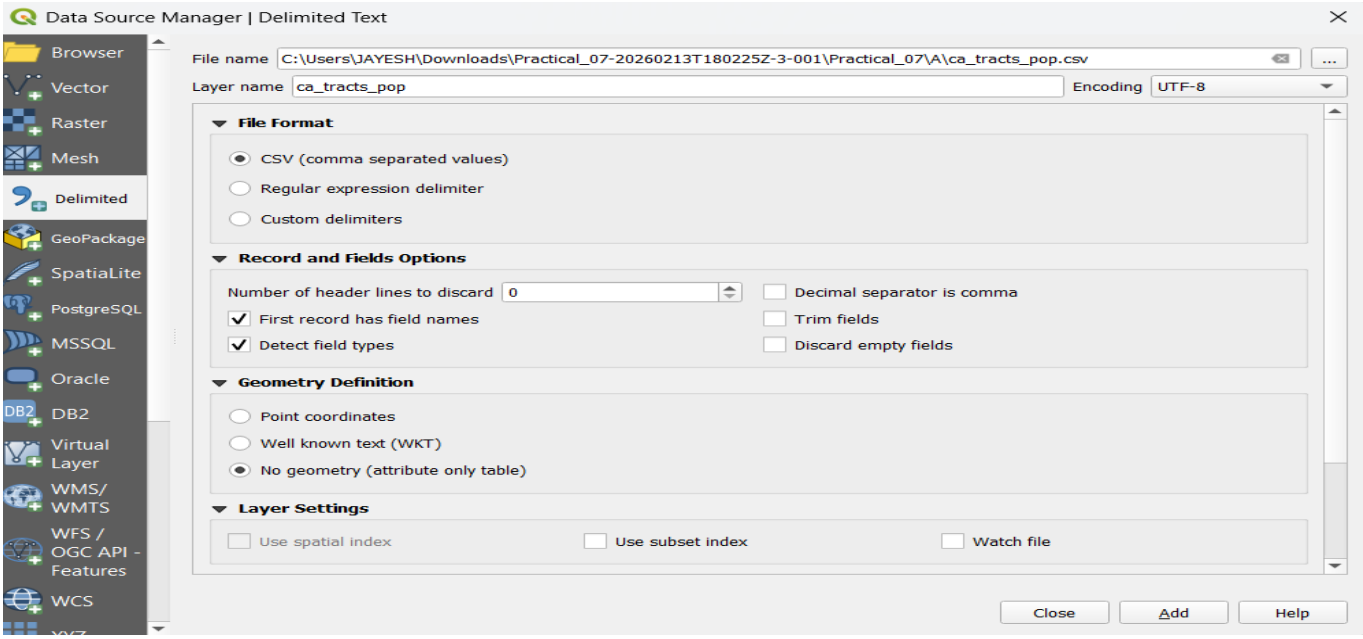
- Start a new project
- Go to Layer→Add Layer→Add new Vector Layer

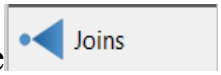
"I:\GIS_Workshop\Practicals\Practical_07\A\Data\t1_2013_06_tract.zip"

- We could import this csv file without any further action and it would be imported. But, the default type of each column would be a string (text).That is ok except for the D001 field which contains numbers for the population. Having those imported as text would not allow us to run any mathematical operations on this column. To tell QGIS to import the field as a number, we need to create a sidecar file with a.csvt extension.

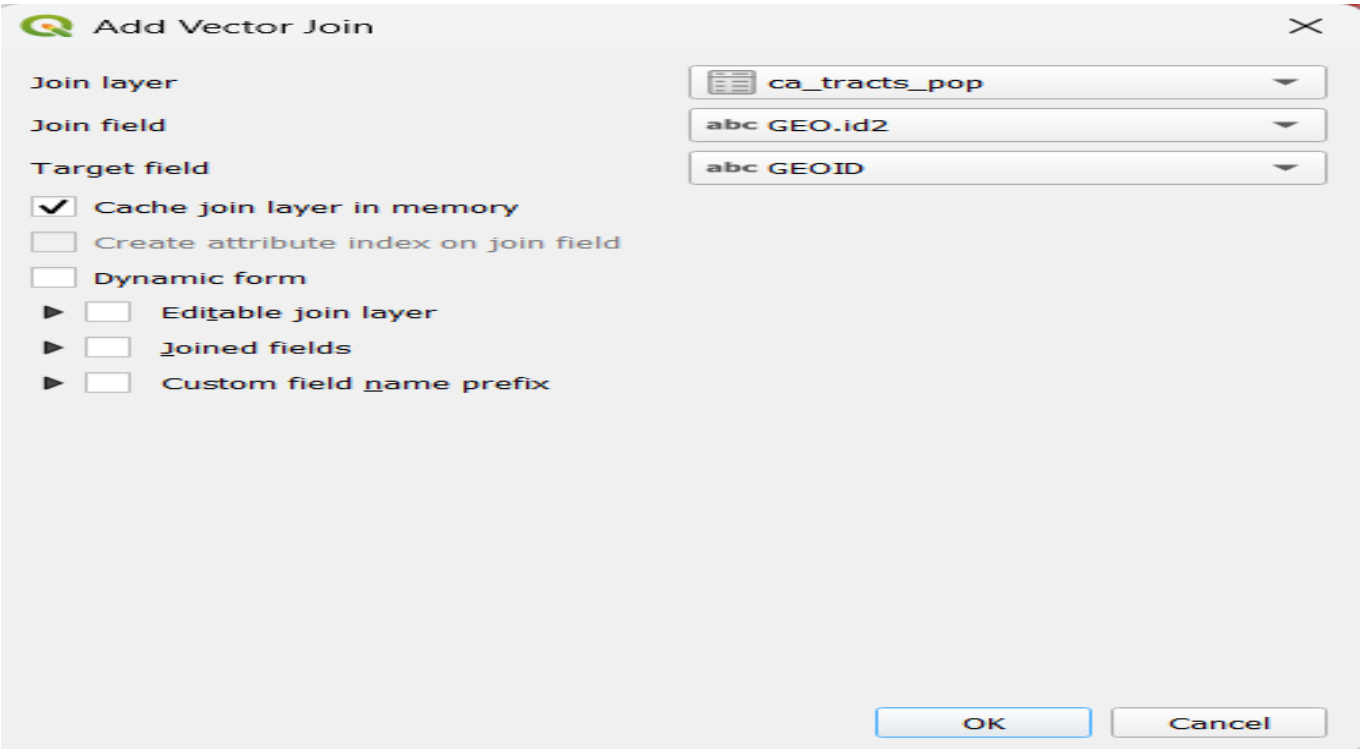


- This file will have only 1 row specifying data types for each column. Save this file as ca_tracts_pop.csvt in the same directory as the original.csv file.
- Go to Layer→Add Layer→Add Delimited Text Layer and add "I:\GIS_Workshop\Practicals\Practical_07\A\Data\ca_tacts_pop.csv"

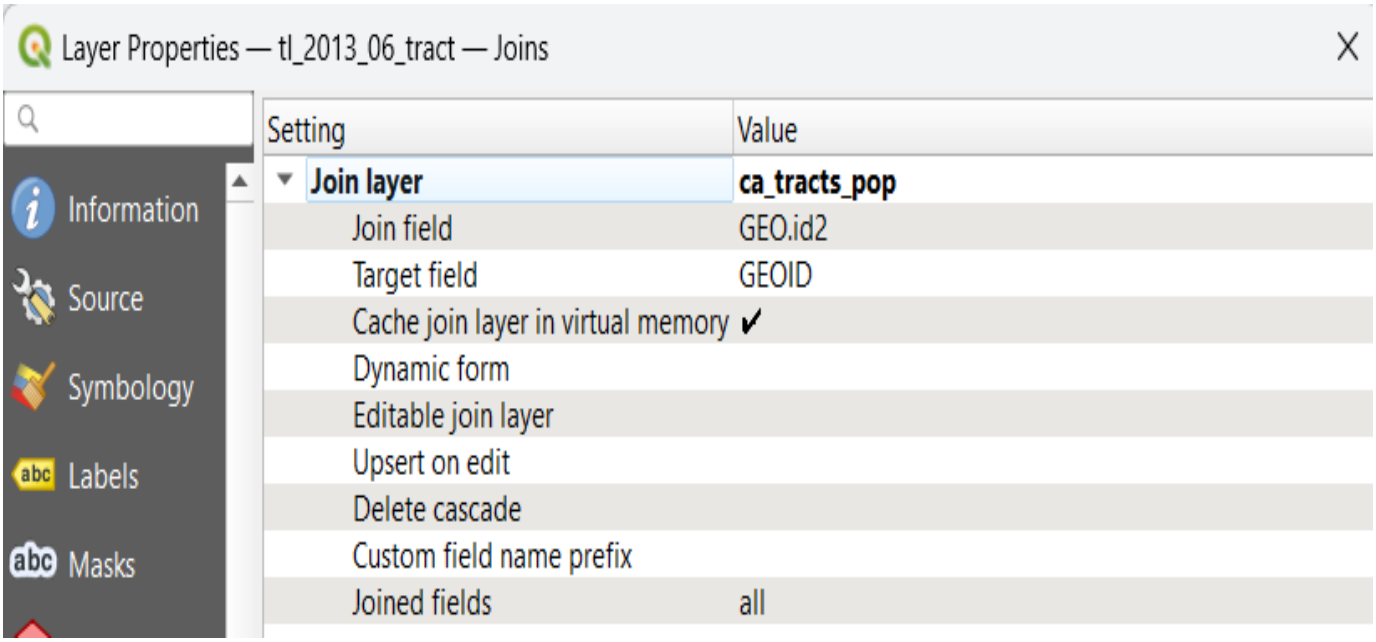


- In the layer panel, right click on “tl_2013_06_tract”, layer and select properties.
- Select the 

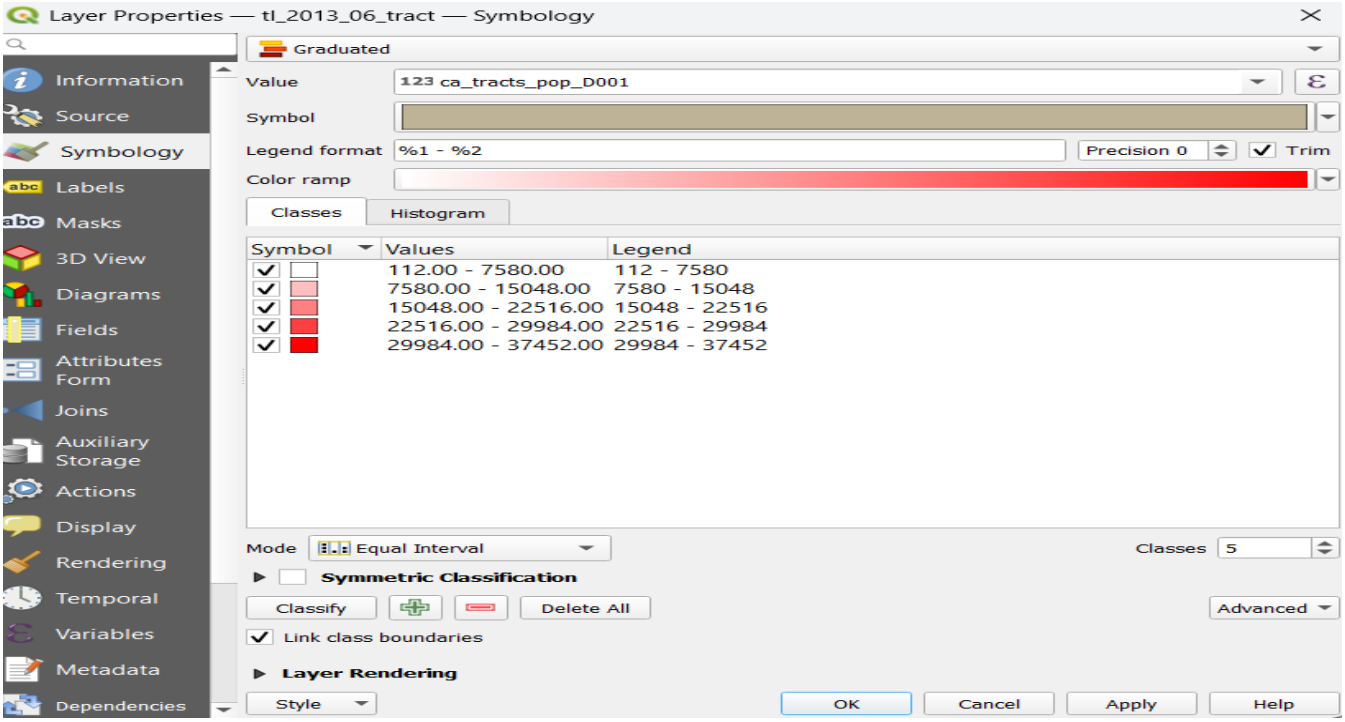
option in properties, and click on button to add new table join.
- In the add vector join windows set the following properties and click OK.



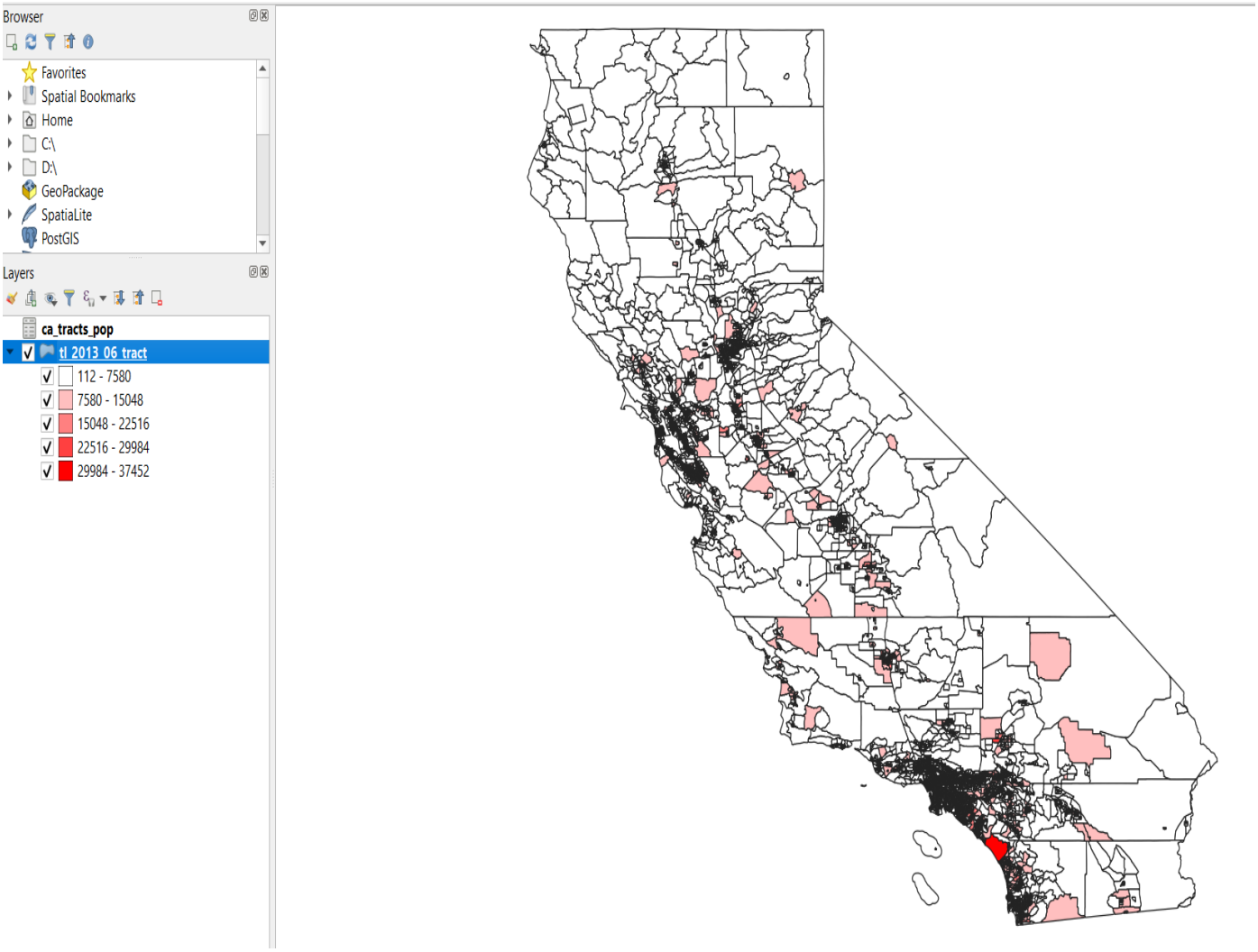
- After performing join



- For more clear output, select “tl_2013_06_tact” from layer panel, right click and select properties. Go to Symbology and set the following properties.

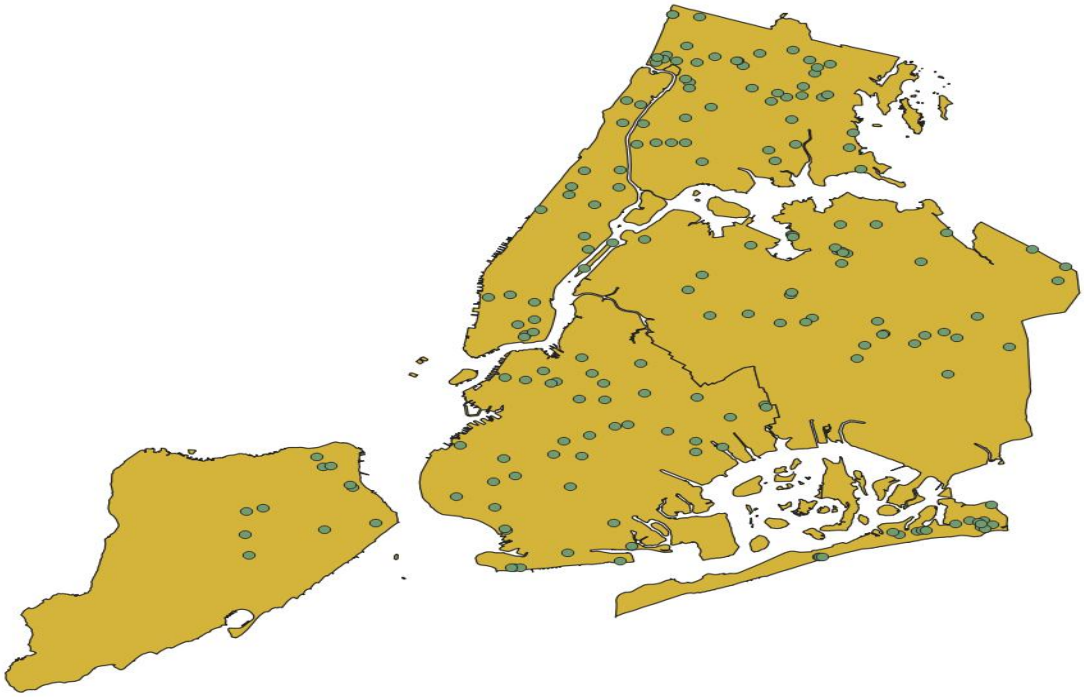


- A detailed and accurate population map of California can be seen as the result. Same technique can be used to create maps based on variety of census data



B. Spatial Joins

- Go to Layer→Add Layer→Add Vector Layer→Select
- **"I:\GIS_Workshop\Practicals\Practical_07\B\Data\nybb_12c\nybb_13c_av\nybb.shp"&"I:\GIS_Workshop\Practicals\Practical_07\B\Data\OEM_NursingHomes_001\OEM_NursingHo mes_001.shp",** from data folder.



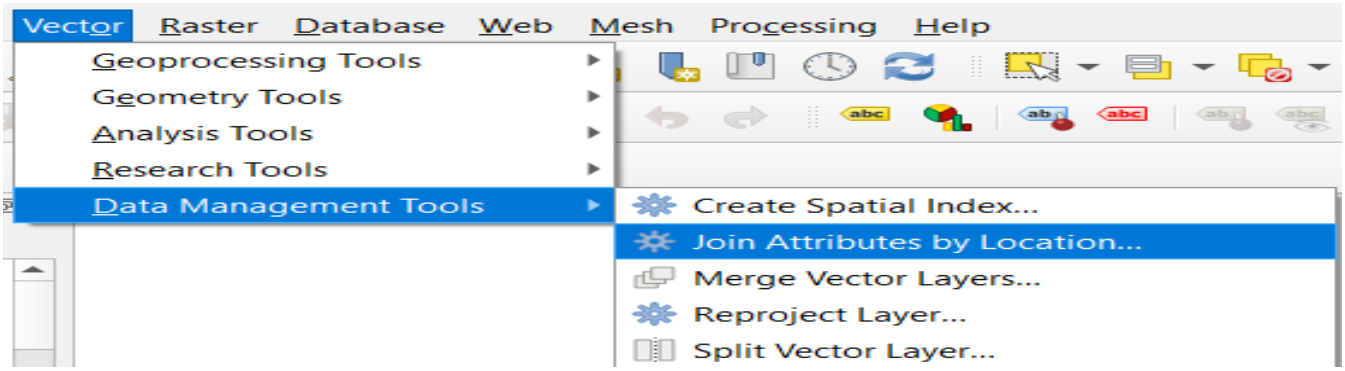
- Go to attribute table and observe the data.
- Table before performing join.

OEM_NursingHomes_001 — Features Total: 177, Filtered: 177, Selected: 0

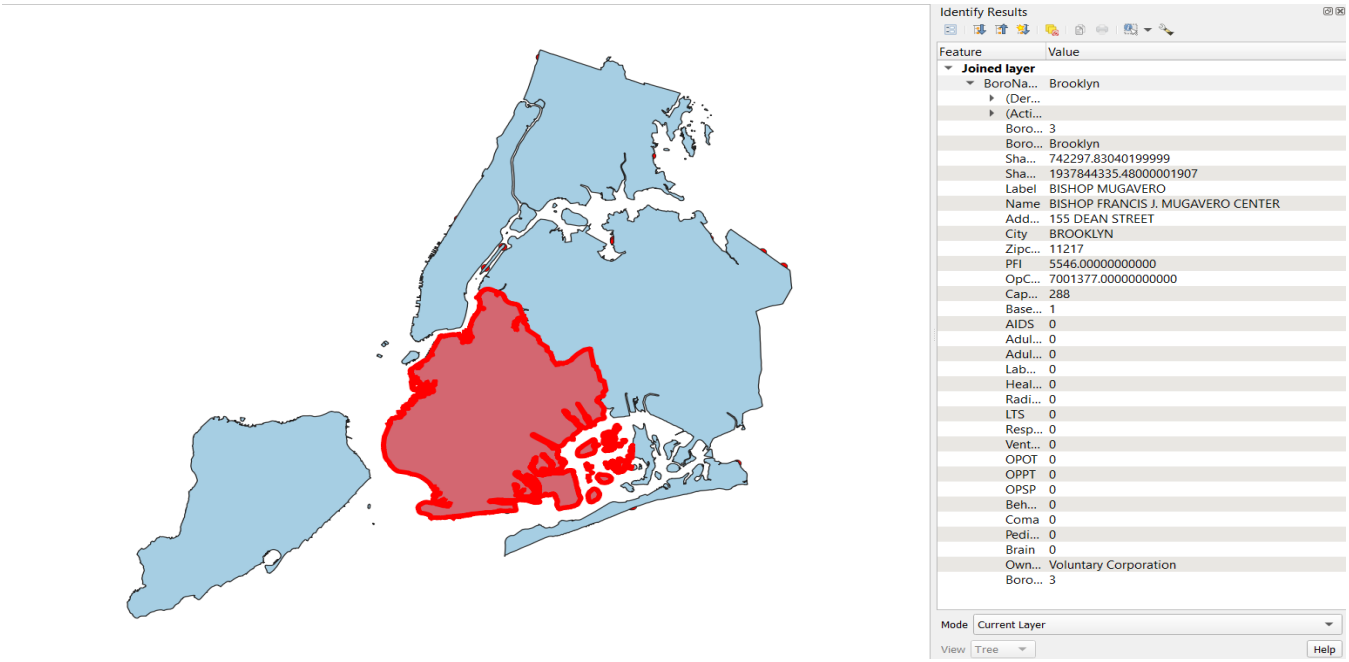
	Label	Name	Address	City	Zipcode	PFI	OpCert	Capacity
1	BROOKLYN ME...	BROOKLYN UNI...	1485 DUMONT ...	BROOKLYN	11208	1368.00000000...	7001308.00000...	120
2	BISHOP HUCLES	BISHOP HENRY ...	835 HERKIMER ...	BROOKLYN	11233	7069.00000000...	7001379.00000...	240
3	BUENA VIDA	BUENA VIDA C...	48 CEDAR STRE...	BROOKLYN	11221	6248.00000000...	7001383.00000...	240
4	BROOKLYN-QU...	BROOKLYN-QU...	2749 LINDEN B...	BROOKLYN	11208	277.00000000000	7001382.00000...	140
5	ROBERT MAPPL...	ROBERT MAPPL...	327 EAST 17TH ...	NEW YORK	10003	4807.00000000...	7002351.00000...	28
6	BISHOP MUGAV...	BISHOP FRANCI...	155 DEAN STRE...	BROOKLYN	11217	5546.00000000...	7001377.00000...	288
7	ATLANTIS	ATLANTIS REH...	140 ST EDWAR...	BROOKLYN	11201	1405.00000000...	7001389.00000...	400
8	NY CRN	NY CENTER FO...	26-13 21ST STR...	ASTORIA	11102	6384.00000000...	7003405.00000...	280
9	FOUR SEASONS	FOUR SEASONS...	1555 ROCKAWA...	BROOKLYN	11236	3227.00000000...	7001385.00000...	270
10	DITMAS PARK	DITMAS PARK C...	2107 DITMAS A...	BROOKLYN	11226	1576.00000000...	7001393.00000...	200
11	HOLY FAMILY	HOLY FAMILY H...	1740 84TH STR...	BROOKLYN	11214	1406.00000000...	7001365.00000...	200
12	HAYM SALOM...	HAYM SALOMO...	2340 CROPSEY ...	BROOKLYN	11214	1361.00000000...	7001369.00000...	240

Show All Features

- Go to Vector→ Data Management Tools→Join Attributes by Location

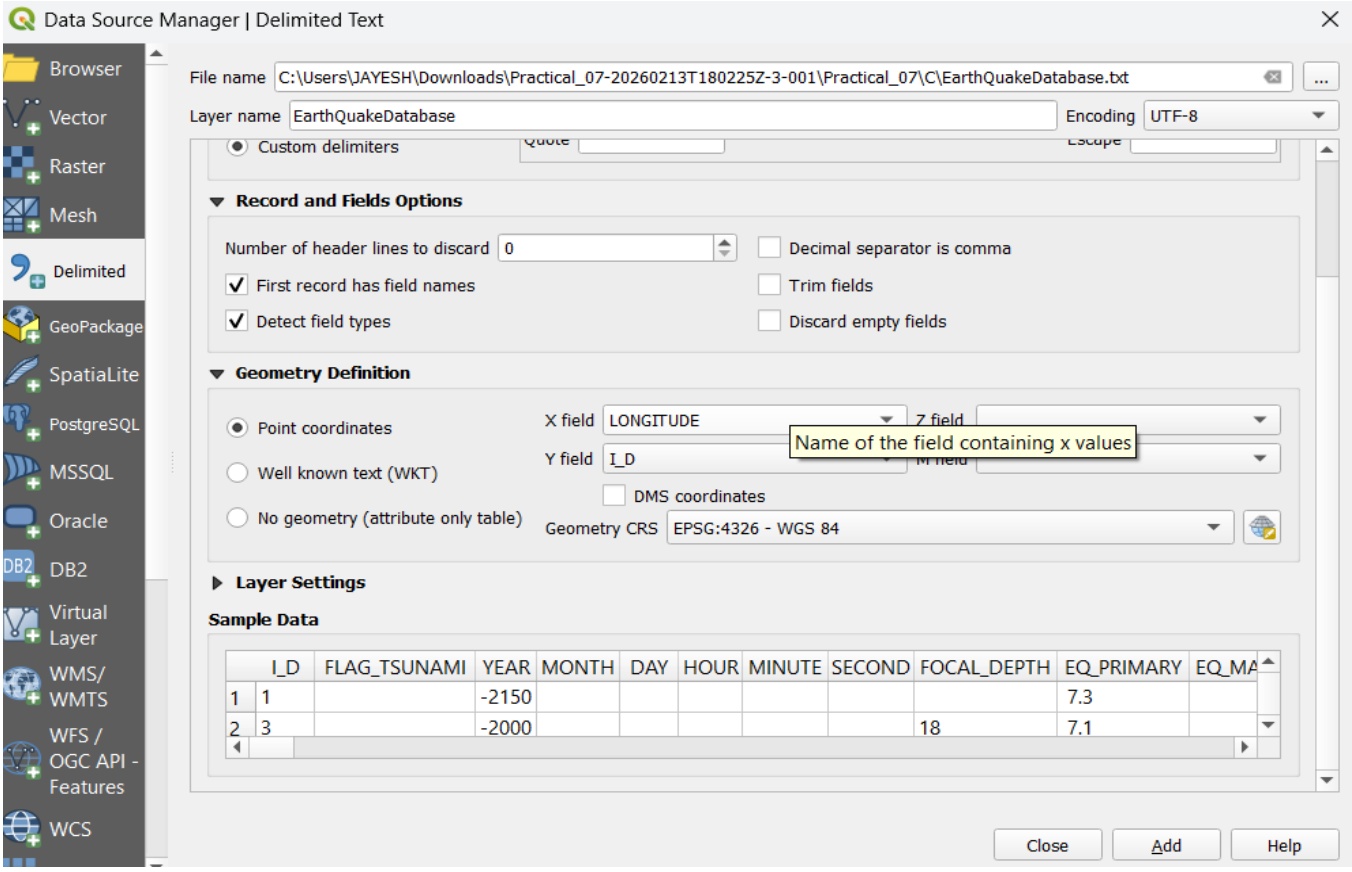


- Attribute table after join
- Use the Identify Feature Button to select a region to view join data on map Layer.
- Output:

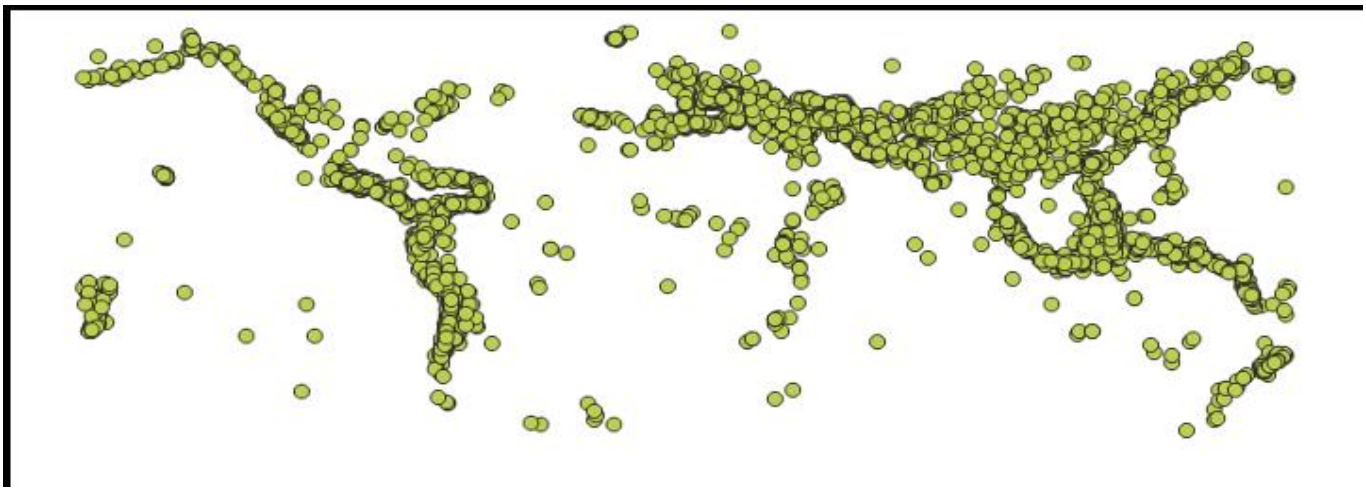


C. Points in Polygon Analysis

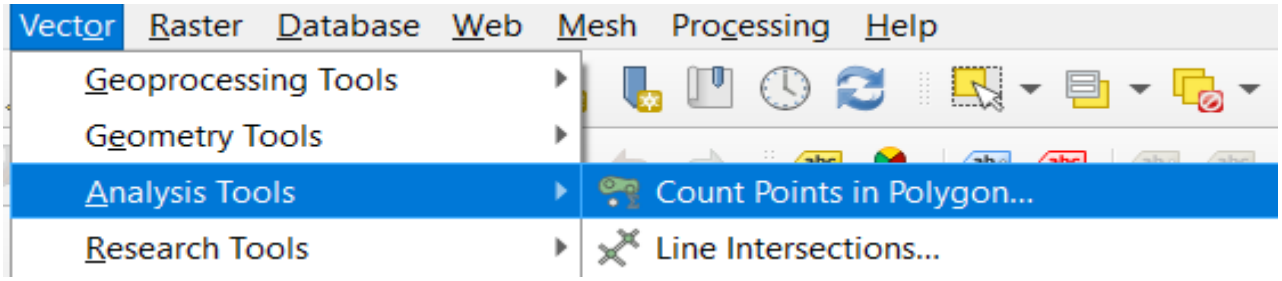
- Go to Layer→Add Layer→Add Delimited Text Layer Select→ "EarthQuakeDatabase.txt"



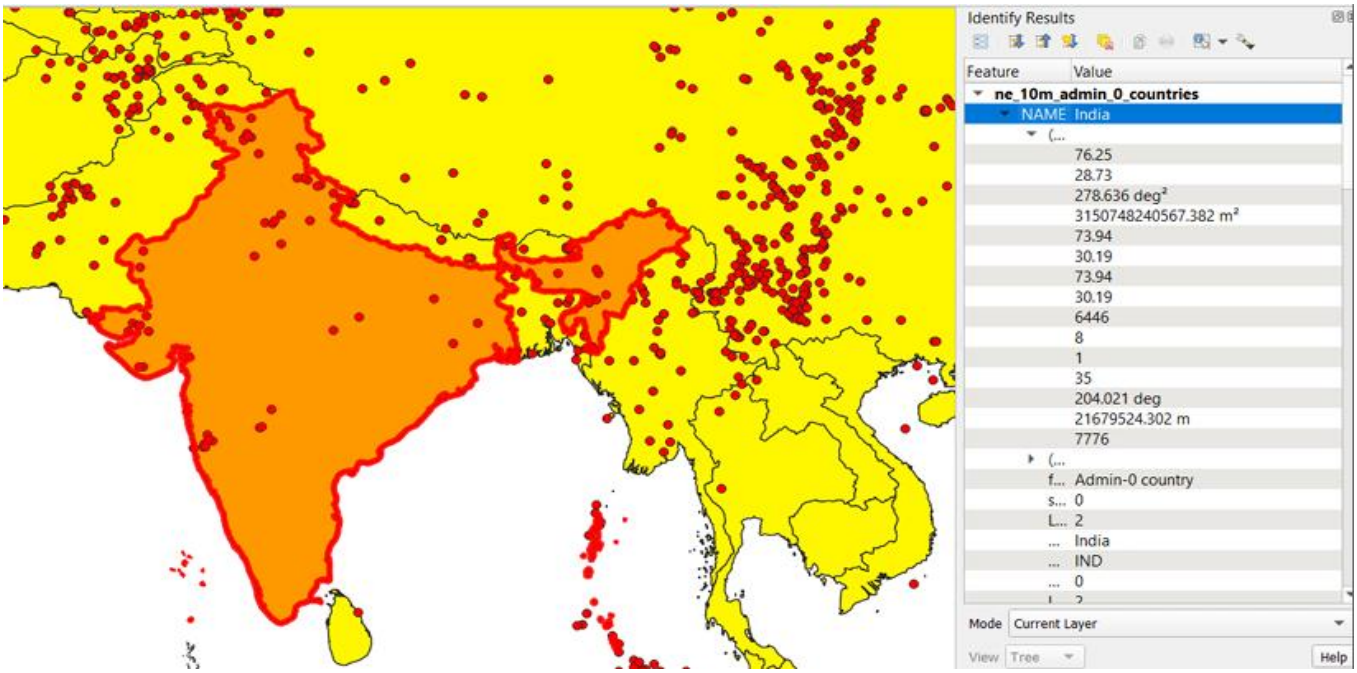
- Go to Layer → Add Layer → Add Delimited Text Layer **"I:\GIS_Workshop\Practicals\Practical_07\C\Data\ne_10m_admin_0_countries.zip"**



- Go to Vector Layer → Select Analysis Tool → Select Count Points in Polygons.



- Use the select Feature button to check country wise counting of Earthquakes.



- Also a new column is added to attribute table “NumPoints” indicating number of earth quake Points in each country.

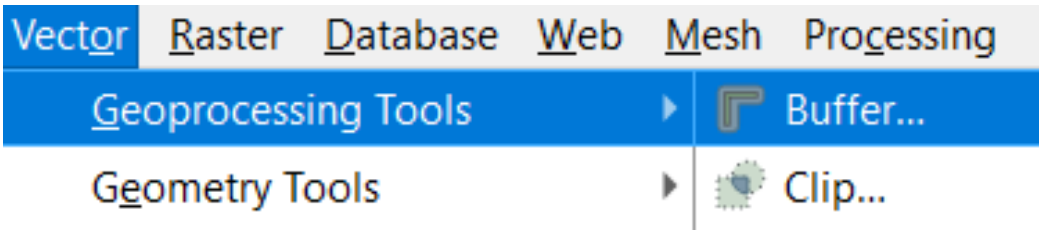
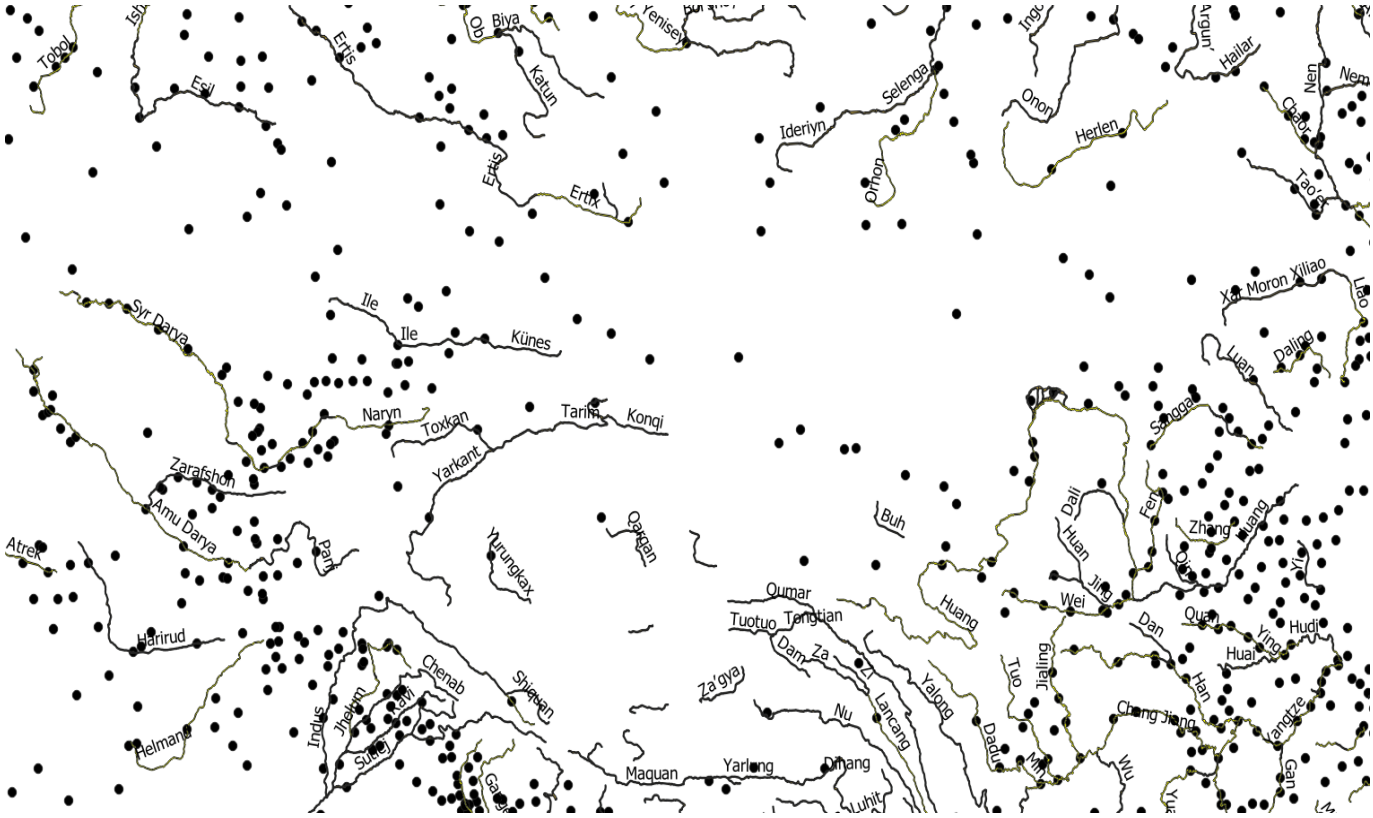
D. Performing Spatial Queries

- Go to Layer→Add Layer→Add vector layer and load
- **“\GIS_Workshop\Practicals\Practical_07\D\Data\ne_10m_populated_places_simple\ne_10m_populated_places_simple.shp”**

“I:\GIS_Workshop\Practicals\Practical_07\D\Data\ne_10m_rivers_lake_centerlines\ne_10m_rivers_lake_centerlines.shp” from project data folder.

- Open project Properties→Set CRS “World_Azimuthal_Equidistant EPSG 54032”.The map will be re-projected as

- Go to Vector→Geoprocessing Tool→Buffer



- Repeat the step to create River Buffer
- Create a buffer for River